

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING



AAT - 22AI643

Report on

Robotic Technology

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CERTIFICATE

This is to certify that the AAT work for the course “Robotic Technology” (Course Code: 22AI643), carried out as part of the AAT component, is submitted by Devansh Goyal (1DS23AI019), Nisha S (1DS23AI041) and Sai Ganesh Ponugoti (1DS23AI042). This report is a record of the AAT (Case Study & Simulation) undertaken by the students as part of their academic curriculum under our supervision. It is submitted towards the partial fulfillment of the requirements for the 6th Semester of the Bachelor of Engineering in Artificial Intelligence and Machine Learning during the academic year 2025–2026. It is certified that all the suggestions and corrections indicated during the internal assessment have been duly incorporated into the report. The AAT report has been approved, as it satisfies the academic requirements in accordance with the rules and regulations prescribed for the Bachelor of Engineering degree.

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1. INTRODUCTION

Robotics is an important field in modern engineering that focuses on the design and development of intelligent machines capable of performing tasks autonomously. One of the major challenges in robotics is enabling a robot to maintain stable navigation and orientation while moving through dynamic environments. Autonomous robots often experience unwanted drift, tilting, or directional errors due to uneven surfaces, wheel mismatch, and environmental disturbances. Therefore, real-time orientation monitoring and motion correction are essential for reliable autonomous navigation.

In this project, an autonomous mobile robot is designed and simulated using the Webots platform. The robot is equipped with an InertialUnit (IMU) sensor, distance sensors, motors, and a display device. The main objective of the project is to continuously monitor the robot orientation in terms of roll, pitch, and yaw while maintaining stable heading during navigation. The system also detects unintended tilts or rotations and automatically corrects robot movement in real time.

The InertialUnit sensor helps the robot determine its orientation accurately by continuously measuring roll, pitch, and yaw values. The robot compares the current yaw angle with the target heading and adjusts motor speeds whenever directional drift occurs. Roll and pitch values are also monitored to detect dangerous tilt conditions. If the robot experiences excessive tilt, the system immediately stops movement to prevent instability or flipping.

To achieve intelligent autonomous behavior, the robot is controlled using a sensor-based feedback control system. Different modules such as heading stabilization, tilt detection, obstacle avoidance, and motion correction work together to control the robot movement. When unstable conditions or obstacles are detected, higher-priority safety behaviors override normal navigation behavior to ensure safe and stable operation.

A display module is also implemented to show real-time roll, pitch, and yaw values along with robot status information during simulation. The project demonstrates how orientation sensing, feedback control, and autonomous correction techniques can be used to develop stable and intelligent robotic systems for applications such as warehouse automation, autonomous vehicles, planetary exploration, and industrial robotics.

1.1 Background and Motivation

Traditional robots operating in fixed and controlled environments are increasingly being replaced by intelligent autonomous systems capable of adapting to dynamic conditions and unexpected disturbances. Modern autonomous robots must maintain stable navigation, detect environmental changes, and make real-time decisions while operating safely in different terrains and environments. One of the major challenges in mobile robotics is maintaining accurate orientation and stable movement during navigation.

Inertial Measurement Unit (IMU) technology plays an important role in solving these challenges by continuously monitoring robot orientation in terms of roll, pitch, and yaw. IMU sensors provide real-time orientation data that helps robots maintain stable heading, detect dangerous tilts, and correct unintended rotations during movement. These sensors are widely used in autonomous vehicles, drones, warehouse robots, planetary rovers, and industrial automation systems because of their high accuracy and fast response.

The motivation behind this project is to demonstrate a practical implementation of IMU-based orientation monitoring and real-time motion correction using the Webots simulation platform. The robot continuously analyzes orientation values from the InertialUnit sensor and dynamically adjusts motor speeds to maintain stable navigation. The system also detects unsafe tilt conditions and performs automatic correction whenever instability or directional drift occurs.

This project reflects the working principles used in real-world autonomous robotic systems deployed in warehouses, industrial automation, autonomous vehicles, and planetary exploration robots, where stable navigation and orientation control are critical for reliable operation.

1.2 Problem Statement

In today's automation-driven industries, there is a growing demand for intelligent robotic systems capable of operating autonomously while maintaining stable and accurate navigation. One of the major challenges in autonomous robotics is enabling a robot to continuously monitor its orientation and maintain proper heading during movement. Robots operating in dynamic environments often experience directional drift, unintended tilts, or instability due to uneven terrain, wheel mismatch, collisions, or external disturbances.

Traditional robotic systems that rely only on wheel movement and predefined control methods may fail to maintain stable navigation under changing environmental conditions. Such instability can reduce navigation accuracy, affect operational safety, and increase the risk of robot failure or collisions.

The problem addressed in this project is to design and develop an autonomous mobile robot in Webots that uses an InertialUnit (IMU) sensor to continuously monitor roll, pitch, and yaw values during navigation. The robot must be capable of maintaining stable heading, detecting unintended tilts or rotations, and correcting its motion automatically in real time using sensor-based feedback control.

Additionally, the robot should detect unsafe tilt conditions, avoid obstacles during movement, and display real-time orientation values and robot status on the screen without human intervention.

1.3 Objectives

- To design a simulation environment using Webots for autonomous robot navigation
- To implement orientation monitoring using the InertialUnit (IMU) sensor
- To continuously measure roll, pitch, and yaw values during robot movement
- To maintain stable heading using real-time yaw correction techniques
- To detect unintended tilts or rotations using roll and pitch monitoring
- To develop a real-time motion correction system for stable navigation
- To implement obstacle detection and avoidance using distance sensors
- To display real-time orientation values and robot status on the screen
- To implement efficient motor control for smooth and stable movement
- To demonstrate autonomous navigation and stabilization in a simulated environment

1.4 Scope of the Project

This project can be further enhanced and extended in several ways:

- Implementing the system on real robotic hardware instead of simulation only
- Integrating advanced sensors such as LiDAR, cameras, GPS, and ultrasonic sensors
- Using AI/ML techniques for intelligent navigation and decision making
- Implementing SLAM (Simultaneous Localization and Mapping) for autonomous mapping and localization
- Adding advanced obstacle detection and terrain analysis systems
- Expanding the system into multi-robot autonomous navigation systems
- Improving navigation stability using advanced sensor fusion techniques
- Developing wireless monitoring and remote robot control systems
- Enhancing motion correction algorithms for rough terrain navigation
- Applying the system in real-world applications such as:
 - Autonomous warehouse robots
 - Automated Guided Vehicles (AGVs) in industries
 - Delivery and service robots
 - Planetary exploration rovers
 - Autonomous surveillance systems
 - Smart transportation systems
- Adding real-time path tracking and navigation optimization features
- Improving orientation accuracy using advanced IMU calibration techniques

2. Case Study: NASA Perseverance Mars Rover

2.1 Introduction

The Perseverance Rover is an advanced autonomous robotic rover developed by NASA for Mars exploration. The rover was launched to explore the Martian surface, collect rock and soil samples, analyze environmental conditions, and send scientific data back to Earth. It is one of the most sophisticated robotic systems developed for space exploration and demonstrates the practical application of autonomous navigation, orientation monitoring, and intelligent robotic control.

The rover operates in extremely harsh and unpredictable terrain where maintaining stability and navigation accuracy is very important. To move safely across rocky surfaces and steep slopes, the rover continuously monitors its orientation using advanced inertial sensing systems. It detects unwanted tilts, maintains stable heading, and corrects its movement automatically while navigating the Martian surface.

The concepts used in the Perseverance Rover are closely related to this project developed in Webots. In this project, the autonomous robot uses an InertialUnit (IMU) sensor to monitor roll, pitch, and yaw values, maintain stable heading, detect dangerous tilts, and perform real-time motion correction during navigation.

2.2 Working Principle

The Perseverance Rover uses multiple sensors, onboard processors, cameras, and navigation systems to move autonomously on the Martian surface. One of the important components used in the rover is the inertial sensing system, which helps monitor orientation and stability during movement.

The rover continuously measures:

- Roll → Side tilt of the rover
- Pitch → Forward/backward tilt
- Yaw → Heading direction

Using these orientation values, the rover adjusts wheel movement and steering to maintain stability and safe navigation across uneven terrain.

Similarly, in this project, the robot uses the InertialUnit sensor in Webots to continuously read roll, pitch, and yaw values. The controller compares the current yaw with the target heading and adjusts motor speeds to maintain stable navigation.

Yaw correction logic:

$$\text{yaw_error} = \text{target_yaw} - \text{current_yaw}$$
$$\text{yaw_error} = \text{target_yaw} - \text{current_yaw}$$

Motion correction:

$$\text{correction} = K \times \text{yaw_error}$$

The robot also detects dangerous tilt conditions using roll and pitch values. If the tilt exceeds the predefined threshold, the robot stops automatically to prevent instability.

2.3 Applications of NASA Perseverance Mars Rover

2.3.1 Planetary Exploration

The primary application of the Perseverance Rover is planetary exploration. The rover studies the Martian surface, atmosphere, rocks, and soil to understand the history of Mars and search for signs of ancient life.



Figure 2.3.1: Mars Rover used in Planetary Exploration

2.3.2 Autonomous Navigation

The rover demonstrates advanced autonomous navigation capabilities by moving safely across rough terrain while maintaining orientation and avoiding obstacles.

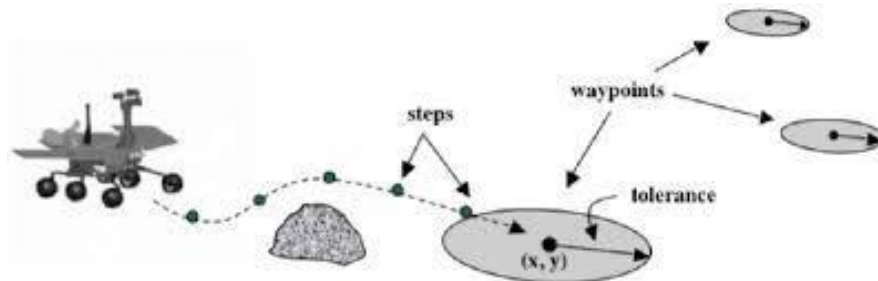


Figure 2.3.2: Autonomous Navigation

2.3.3 Scientific Data Collection

The rover collects samples and sends scientific data back to Earth for research and analysis.

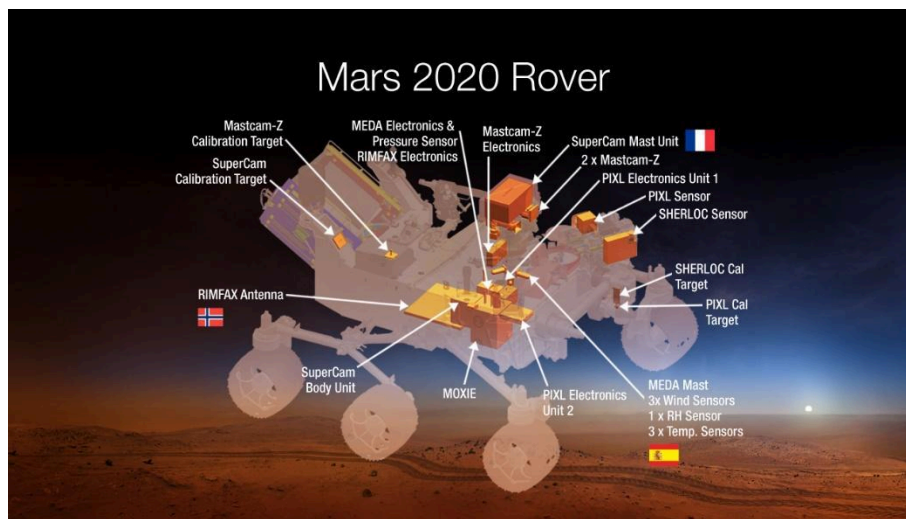


Figure 2.3.2: Scientific Data Collection tools in Rover.

2.4 Advantages

The technologies demonstrated by the Perseverance Rover and this project provide several advantages:

- Improved navigation stability
- Real-time orientation monitoring
- Automatic motion correction

-
- Safer autonomous movement
 - Better obstacle handling
 - Reliable operation in dynamic environments

2.5 Limitations of the Da Vinci Surgical System

Despite their advantages, such robotic systems also have limitations:

- Sensor noise may affect accuracy
- Harsh environments can impact sensor performance
- Complex terrain requires advanced navigation algorithms
- Autonomous systems require high computational power

2.6 Real World Application

2.6.1 Perseverance Rover in Mars Exploration

One of the most significant real-world applications of the Perseverance Rover is autonomous planetary exploration on Mars. The rover is used by NASA to explore the Martian surface, collect rock and soil samples, analyze environmental conditions, and send scientific data back to Earth. The rover uses advanced orientation monitoring systems and autonomous navigation technologies to move safely across rough and uneven terrain. Its ability to detect tilt, maintain stable heading, and correct movement in real time helps prevent instability while navigating the challenging Martian environment.



Figure 2.6.1: Perseverance Rover

2.6.2 Autonomous Navigation and Stability Control in Space Robotics

Another important real-world application of technologies similar to this project is autonomous navigation and stability control in space robotics. Space exploration robots and planetary rovers use inertial sensing systems such as IMU sensors to continuously monitor roll, pitch, and yaw during movement. These systems help maintain balance, avoid dangerous tilts, and ensure safe navigation on rocky or sloped surfaces. Autonomous stabilization systems are also used in lunar rovers, unmanned exploration vehicles, and research robots operating in extreme environments where human control is limited.

2.7 Future Scope of the Perseverance Rover

The future of the Perseverance Rover is expected to be driven by rapid advancements in robotics, artificial intelligence, autonomous navigation, and space exploration technologies. One of the most promising developments is the integration of advanced artificial intelligence systems that can help future rovers make more intelligent navigation and scientific decisions without direct human intervention. AI-based systems may allow rovers to identify terrain conditions, detect obstacles more accurately, and optimize exploration paths automatically.

Another important area of development is the improvement of autonomous navigation and orientation stabilization systems. Future Mars rovers are expected to use more advanced IMU sensors, terrain mapping systems, and sensor fusion techniques to improve movement accuracy and stability on rough planetary surfaces. These technologies will help robots maintain balance, detect dangerous tilts, and perform safer navigation in extreme environments.

Researchers are also focusing on improving communication technologies for deep-space exploration. Future rovers may support faster and more reliable transmission of scientific data, images, and environmental information back to Earth. Advanced robotic systems may also enable remote robotic operations and cooperative exploration using multiple autonomous rovers working together.

Future developments may include:

- AI-based autonomous decision-making systems
- Advanced terrain mapping and obstacle detection
- Improved orientation monitoring and stabilization systems
- Sensor fusion using IMU, LiDAR, cameras, and GPS-like systems
- Multi-rover cooperative exploration systems
- Enhanced robotic arms for scientific sample collection
- Better power management and longer operational lifespan
- Integration of augmented reality and real-time 3D mapping technologies

As space robotics technology continues to evolve, future Mars rovers are expected to become more intelligent, efficient, and autonomous, helping scientists explore distant planets more safely and effectively.

2.8 Conclusion

In conclusion, the Perseverance Rover represents one of the most advanced autonomous robotic systems developed for planetary exploration. Its ability to maintain stable navigation, monitor orientation, detect dangerous tilts, and perform autonomous movement correction has made it highly effective for exploring the Martian surface under extreme environmental conditions.

The rover demonstrates how advanced robotic technologies, sensor systems, and autonomous control mechanisms can work together to achieve reliable and intelligent navigation. Its

successful operation on Mars highlights the importance of orientation monitoring systems such as IMU sensors in maintaining stability and ensuring safe robotic movement on uneven terrain.

Although Mars exploration systems have challenges such as harsh environmental conditions, communication delays, and complex terrain navigation, continuous advancements in robotics, artificial intelligence, and autonomous systems are improving the capabilities of planetary rovers. Future exploration robots are expected to become more intelligent, autonomous, and efficient in performing scientific missions.

Overall, the Perseverance Rover demonstrates the practical application of autonomous robotics, orientation monitoring, and real-time motion correction technologies, which are closely related to the concepts implemented in this project developed using Webots.

3.APPLICATIONS

The developed system is an autonomous robot capable of monitoring orientation, maintaining stable heading, detecting unintended tilts, and correcting motion using sensor-based feedback control. The project is implemented in the Webots simulation environment using a mobile robot equipped with an InertialUnit (IMU) sensor, distance sensors, motors, and a display device. The robot continuously processes orientation data and adjusts motor speeds to ensure stable and safe navigation. This system demonstrates how intelligent robots can operate autonomously while maintaining balance and directional stability in dynamic environments.

In real-world scenarios, similar systems are widely used in industrial automation and autonomous robotics. Automated Guided Vehicles (AGVs) and Autonomous Mobile Robots (AMRs) use orientation sensors and motion correction systems to maintain accurate navigation while transporting materials in factories and warehouses. These robots continuously monitor their heading and correct movement errors to ensure efficient and safe operation. This helps reduce manual labor, improve navigation accuracy, and increase productivity in industries.

Another important application is in warehouse and logistics systems, where autonomous robots are used for sorting, transporting, and delivering goods. IMU-based stabilization systems help robots maintain stable movement while navigating crowded or dynamic environments. Similar technologies are also used in delivery robots and service robots operating in hospitals, offices, airports, and public spaces where autonomous navigation and obstacle avoidance are essential.

The concepts implemented in this project are also highly relevant in autonomous vehicles, drones, and planetary exploration systems. Self-driving vehicles use IMU sensors, orientation monitoring, and real-time motion correction to maintain stable navigation and improve safety during movement. Planetary rovers such as the Perseverance Rover use similar orientation sensing systems to maintain stability while exploring rough terrain surfaces.

Thus, this project provides a practical understanding of real-time orientation monitoring, autonomous stabilization, and intelligent motion correction systems used in modern robotics applications.

Real world applications of robots:

- Industrial Automation – Automated Guided Vehicles (AGVs) in factories
- Warehouse Logistics – Autonomous robots for goods transportation

-
- Healthcare – Medicine delivery and service robots in hospitals
 - Autonomous Vehicles – Orientation monitoring and navigation systems
 - Planetary Exploration – Rover stabilization and navigation systems
 - Service Robots – Cleaning, delivery, and assistance robots
 - Surveillance Systems – Autonomous monitoring robots
 - Smart Transportation – Intelligent navigation and motion control systems

3.1 Industrial Warehouse Automation

Autonomous Mobile Robots (AMRs) are widely used in warehouses and manufacturing industries for efficient material handling, transportation, and inventory management. In real-world systems, these robots use orientation monitoring sensors, navigation systems, and obstacle detection technologies to move safely and accurately within industrial environments.

Companies such as [Amazon Robotics](#), [DHL](#), and [Flipkart](#) use autonomous robots for:

- Material and goods transportation within warehouses
- Safe navigation in environments shared with human workers
- Maintaining stable and accurate movement during operation
- Detecting and avoiding obstacles such as boxes, pallets, and moving objects
- Improving operational efficiency and reducing manual labor

The behavior modeled in this project — monitoring roll, pitch, and yaw using an InertialUnit (IMU) sensor while maintaining stable heading and avoiding obstacles — represents a simplified version of real-world warehouse AMRs. Similar systems are used in industrial robots to ensure stable navigation, orientation correction, and safe autonomous movement in dynamic environments.

3.2 Autonomous Mobile Robots and Navigation Systems

Autonomous robots rely on sensor-based navigation systems to move independently without human intervention. These robots use various sensors such as IMU sensors, distance sensors, cameras, LiDAR, and ultrasonic sensors to monitor orientation, detect obstacles, and make navigation decisions in real time.

Key functionalities include:

- Continuous orientation monitoring using IMU sensors
- Stable heading maintenance during movement
- Detection of nearby obstacles using distance sensors
- Real-time motion correction and navigation control
- Autonomous decision-making based on sensor inputs

This project demonstrates these principles using an InertialUnit (IMU) sensor for monitoring roll, pitch, and yaw values, along with distance sensors for obstacle detection and avoidance. The robot continuously analyzes orientation data and adjusts motor speeds to maintain stable navigation. The implemented system represents a basic model of real autonomous navigation and stabilization systems used in modern mobile robots and autonomous vehicles.

3.3 Industrial and Smart Transportation Systems

In smart transportation and industrial automation systems, autonomous robots and vehicles are required to maintain stable navigation, follow predefined routes, and avoid collisions while operating with minimal human intervention. These systems rely heavily on orientation monitoring, sensor-based navigation, and real-time motion correction to ensure safe and efficient movement.

Important features include:

- Stable heading and orientation maintenance during navigation
- Detection and avoidance of obstacles in the movement path
- Real-time motion correction using sensor feedback
- Continuous autonomous movement with minimal human intervention
- Accurate navigation in dynamic environments

In this project, the robot continuously monitors roll, pitch, and yaw values using the InertialUnit (IMU) sensor while maintaining stable heading during movement. Distance sensors are used for obstacle detection and avoidance, while motion correction algorithms adjust motor speeds whenever directional drift occurs. This represents a simplified model of autonomous transportation and industrial robotic systems used in smart factories, autonomous delivery systems, and intelligent transportation technologies.

3.4 Robotics Education and Learning

Simulation tools such as Webots are widely used in educational institutions and research laboratories to teach robotics concepts without requiring expensive physical hardware. These platforms allow students and researchers to design, test, and analyze autonomous robotic systems safely within a virtual environment.

Applications include:

- Learning sensor-based feedback control systems
- Understanding robot orientation and navigation logic
- Studying real-time motion correction techniques
- Testing autonomous algorithms in a safe simulation environment
- Analyzing robot behavior under different environmental conditions

This project serves as a practical example for students to understand IMU-based orientation monitoring, heading stabilization, obstacle detection, and real-time motion correction using simulation. The project also demonstrates how autonomous robots use roll, pitch, and yaw values to maintain stable navigation and perform intelligent decision-making in dynamic environments.

3.5 Research and Development in Robotics

Simulation platforms play an important role in robotics research and development by allowing engineers and researchers to design, test, and improve robotic algorithms before implementing

them in real-world systems. Platforms such as Webots provide a safe and cost-effective environment for analyzing autonomous robot behavior under different operating conditions.

They are widely used for:

- Developing and testing autonomous navigation algorithms
- Studying orientation monitoring and motion correction systems
- Analyzing robot behavior in controlled simulation environments
- Testing obstacle detection and avoidance mechanisms
- Improving navigation efficiency and system reliability
- Reducing hardware development and testing costs

This project demonstrates how simulation can be effectively used to design and validate IMU-based autonomous navigation systems. The robot continuously monitors roll, pitch, and yaw values using the InertialUnit (IMU) sensor, performs real-time motion correction, and maintains stable heading during navigation. The project highlights the importance of simulation in developing reliable autonomous robotic systems before real-world implementation.

4. SIMULATION DESIGN

The simulation for this project is developed using the Webots robotics simulation platform, which provides a virtual environment for designing and testing autonomous robot behavior. The simulation consists of three major components: environment setup, robot design, and control system implementation.

4.1 Environment Design

The simulation environment is created using a rectangular arena that defines the operational area of the robot. Obstacles are added inside the arena using Solid nodes with box geometry to test obstacle detection and avoidance capabilities. Physics properties are enabled to create realistic robot interaction within the environment.

The arena floor is designed to provide stable movement conditions while allowing the robot to experience directional drift and orientation changes during motion. This helps evaluate the performance of the IMU-based stabilization system.

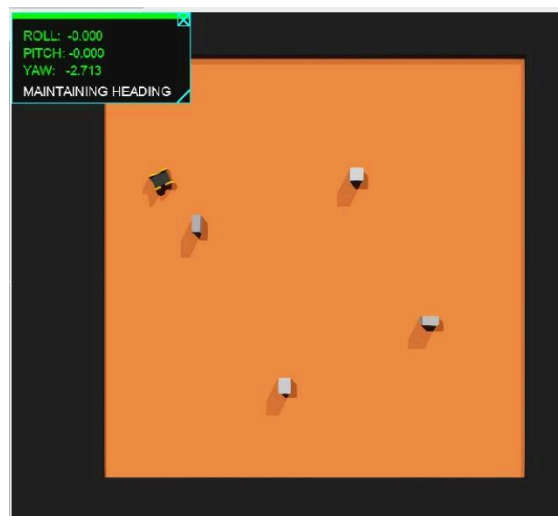


Fig 4.1 Environment Design

4.1.1 Environment Specifications

Parameter	Value	Description
Arena Dimensions	10m × 10m	Rectangular bounded workspace
Floor Texture	Checkerboard	Visual reference for robot position
Small Obstacles	Cylindrical cones (r=0.1m)	Triggers 'continue moving' behavior
Large Obstacles	Rectangular boxes (0.5×0.5×1m)	Triggers 'stop' behavior
Boundary Walls	Height: 1.5m, Thickness: 0.1m	Prevents robot from exiting
Gravity	9.81 m/s ²	Standard Earth gravity
Time Step	32 ms	Simulation timestep

4.2 Robot Model

The robot is designed as a four-wheel differential drive mobile robot. The chassis is created using box geometry, and wheels are connected using hinge joints. Proper wheel anchor positions are configured to ensure smooth wheel rotation without wobbling.

The robot includes the following components:

InertialUnit (IMU)

The IMU sensor is placed near the center of mass of the robot to obtain accurate orientation values. It continuously measures:

- Roll
- Pitch
- Yaw

Distance Sensors

Two infrared distance sensors are mounted at the front-left and front-right positions to detect nearby obstacles.

Motors

Two motors control the left and right wheel movement using differential drive control.

Display Device

A display screen is attached to the robot to show real-time orientation values and robot status.

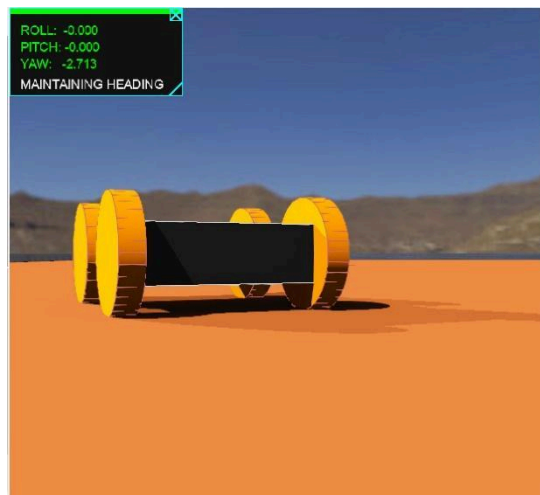


Fig 4.2 Robot design

4.2.1 Robot Physical Specifications

Component	Specification	Purpose
Chassis	Rectangular, $0.3\text{m} \times 0.2\text{m} \times 0.1\text{m}$	Houses electronics and sensors
Drive Wheels	$2 \times$ differential, radius 0.06m	Locomotion control
Caster Wheel	$1 \times$ passive front	Balance and stability
Mass	2.5 kg	Realistic inertia behavior
Max Wheel Velocity	10 rad/s	Speed limiter

Display Panel	$64 \times 32\text{ px}$, front-mounted	Status output
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4.3 Control System Design

The control system is implemented using Python programming inside Webots. The robot operates in a continuous loop where sensor data is read and processed to control motor movement.

The system includes the following modules:

- **Heading Stabilization Module**

This module continuously compares the current yaw angle with the target yaw angle and adjusts wheel speeds to maintain stable heading.

- **Tilt Detection Module**

This module monitors roll and pitch values. If the tilt exceeds a predefined threshold, the robot stops immediately to prevent flipping.

- **Obstacle Avoidance Module**

This module uses distance sensor values to detect obstacles and perform turning actions to avoid collisions.

- **Display Module**

This module updates the screen with live roll, pitch, yaw values, and robot safety status.

4.4 Working of Simulation

The working of the simulation is carried out in the Webots environment using an autonomous mobile robot equipped with an InertialUnit (IMU) sensor, distance sensors, motors, and a display device. The system is designed to enable autonomous navigation by combining orientation monitoring, heading stabilization, obstacle detection, and real-time motion correction behaviors.

1. Initialization Phase

At the start of the simulation, the robot and all its components are initialized using Webots API functions. The motors are configured in velocity control mode, and all sensors are enabled using a fixed simulation time step.

- The InertialUnit (IMU) sensor is activated to measure roll, pitch, and yaw values
- Distance sensors are enabled to detect nearby obstacles
- Motors are initialized for continuous rotation and differential drive control
- The display device is activated to show real-time orientation values and robot status

This ensures that the robot is ready to sense the environment, monitor orientation, and navigate autonomously.

2. Simulation Environment Setup

The simulation environment consists of:

- A rectangular arena acting as the workspace
- Obstacles placed inside the environment
- A mobile robot positioned at the starting point
- A display module for monitoring robot orientation

The robot must maintain stable heading, avoid obstacles, detect dangerous tilts, and perform autonomous motion correction during navigation.

3. Continuous Control Loop

The robot operates inside a continuous control loop, where the following steps are repeated at every simulation time step.

a) Sensor Data Acquisition

The robot continuously reads sensor values in real time.

- The IMU sensor measures:
 - Roll
 - Pitch
 - Yaw
- Distance sensors detect nearby obstacles

The sensor values are continuously updated to reflect real-time robot orientation and environmental conditions.

b) Orientation Monitoring and Heading Stabilization

The robot continuously compares the current yaw value with the target heading.

- If the yaw error is small → robot moves straight
- If the robot drifts left or right → motor speeds are adjusted automatically
- Motion correction is applied to restore stable heading

This ensures accurate and stable navigation throughout the simulation.

c) Tilt Detection and Safety Control

The robot continuously monitors roll and pitch values to detect instability.

- If roll or pitch exceeds the safety threshold:
 - The robot identifies an unsafe tilt condition
 - All movement is stopped immediately
 - Safety behavior overrides all other navigation behaviors

This prevents the robot from flipping or losing stability during operation.

d) Obstacle Detection and Avoidance

When an obstacle is detected using distance sensors:

- The robot identifies the obstacle direction
- Obstacle avoidance behavior overrides normal heading stabilization
- The robot performs turning or pivot correction to avoid collision

This allows safe autonomous movement inside the environment.

e) Motion Correction Behavior

The robot continuously performs real-time motion correction using IMU feedback.

- Motor speeds are adjusted dynamically based on orientation values
- The robot maintains balanced and stable movement
- Unwanted rotations and directional drift are corrected automatically

This behavior ensures smooth autonomous navigation.

4. Motor Control

The robot uses differential drive motion:

- Left and right motor speeds are controlled independently
- Speed values are adjusted using orientation and obstacle sensor inputs

Examples:

- Straight movement → both motors rotate at equal speed
- Heading correction → one motor rotates faster than the other
- Obstacle avoidance → opposite motor directions for turning
- Tilt detection → both motors stop immediately

5. Output and Debugging

The display module and simulation console show real-time information such as:

- Roll, pitch, and yaw values
- Robot heading status
- Tilt detection warnings
- Obstacle detection status
- Active control behaviors
- Robot motion state information

This helps in debugging, monitoring robot performance, and understanding the autonomous navigation behavior of the system.

4.5 Webots World File Structure

The Webots world (.wbt) file defines all simulation elements in a scene tree format. Key nodes include:

- WorldInfo node: Gravity, timestep, and physics settings.
- Viewpoint node: Camera position and orientation for the simulation window.
- Robot node: Contains the robot's body, wheels, LiDAR, and Display as child nodes.
- Solid nodes: Static obstacle objects with defined shapes, positions, and materials.

5. SDG MAPPING

The developed project of an autonomous IMU-based robot aligns with several United Nations Sustainable Development Goals (SDGs) because it promotes intelligent automation, innovation, and safe robotic navigation systems.

SDG 9 – Industry, Innovation, and Infrastructure

This project strongly supports SDG 9 by promoting innovation in robotics, autonomous systems, and sensor-based navigation technologies. Orientation monitoring and real-time motion correction are essential in modern industrial automation systems. Autonomous robots with stabilization mechanisms improve efficiency, reliability, and safety in industrial environments.

SDG 11 – Sustainable Cities and Communities

The concepts used in this project are applicable to smart city systems such as autonomous delivery robots, service robots, and intelligent transportation systems. Stable navigation and obstacle avoidance contribute to safer and more efficient urban automation solutions.

SDG 8 – Decent Work and Economic Growth

The implementation of automation technologies like this project helps improve productivity and efficiency in industries. By reducing repetitive manual tasks and enhancing operational accuracy, such systems support economic growth. At the same time, they encourage the development of skilled jobs in robotics, programming, and system design, contributing to technological advancement and employment opportunities.

6. IMPLEMENTATION

The implementation of the Autonomous Rover simulation is carried out using the Webots simulation platform with a Python-based controller. The system is designed to enable the autonomous rover to navigate stable headings, detect and avoid obstacles, monitor real-time 3D orientation (Roll, Pitch, Yaw), perform real-time motion correction, and render telemetry on an onboard display.

6.1 Development Environment Setup

6.1.1 Software Requirements

Software	Version	Purpose
Webots	R2023b or later	Robotic simulation environment
Python	3.10+	Robot controller language
NumPy	1.24+	LiDAR array processing
Webots Python API	Bundled with Webots	Robot device interface
OS	Windows 10/11, Ubuntu 20.04+, macOS 12+	Simulation host

6.2 World File Configuration (.wbt)

The simulation scene and environment are defined inside the Webots world file ('Autonomous_Robot.wbt'). It contains world configurations, lighting, arena dimensions, solid obstacles, and the rover's node representation.

6.2.1 Robot Node Definition

```
Robot {  
  translation -2.74 2.20 0.09  
  children [  
    InertialUnit {  
      name "imu"  
    }  
    Display {  
      width 256  
      height 128  
    }  
    DistanceSensor {  
      name "ds_left"  
    }  
    DistanceSensor {  
      name "ds_right"  
    }  
    HingeJoint { # Left wheel joint  
      ..  
    }  
    HingeJoint { # Right wheel joint  
      ..  
    }  
  ]  
  controller "imu_controller"  
}}
```

6.3 Python Controller: Full Logic

The controller file acts as the onboard computer (on-board processor) of the rover. It processes sensory data and issues motor speed commands in a continuous feedback loop.

6.3.1 Initialization

```
robot = Robot()
time_step = int(robot.getBasicTimeStep())

# Initialize motors
left_motor = robot.getDevice('left_motor')
right_motor = robot.getDevice('right_motor')
left_rear_motor = robot.getDevice('left_rear_motor')
right_rear_motor = robot.getDevice('right_rear_motor')

for motor in [left_motor, right_motor, left_rear_motor, right_rear_motor]:
    motor.setPosition(float('inf'))
    motor.setVelocity(0.0)

# Initialize sensors
imu = robot.getDevice('imu')
imu.enable(time_step)

ds_left = robot.getDevice('ds_left')
ds_right = robot.getDevice('ds_right')
ds_left.enable(time_step)
ds_right.enable(time_step)

display = robot.getDevice('display')
```

6.3.2 Reading sensors

```
while robot.step(time_step) != -1:
    # Read orientation from IMU
    rpy = imu.getRollPitchYaw()
    roll, pitch, yaw = rpy[0], rpy[1], rpy[2]

    # Read front proximity sensors
    left_val = ds_left.getValue()
    right_val = ds_right.getValue()

    left_speed = 5.0
    right_speed = 5.0
```

6.3.3 Obstacle Avoidance

```
if left_val > 100 or right_val > 100:
    # turn away from obstacle
    left_speed = -5.0
    right_speed = 5.0
```

6.3.4 Main Control Loop

```
# Tilt Safety Check
if abs(roll) > 0.2 or abs(pitch) > 0.2:
    left_speed = 0.0
    right_speed = 0.0
# Heading Stabilization
elif left_val <= 100 and right_val <= 100:
    yaw_error = target_yaw - yaw
    correction = 2.0 * yaw_error
    left_speed = 5.0 - correction
    right_speed = 5.0 + correction

# Apply speeds
```

```
left_motor.setVelocity(left_speed)

right_motor.setVelocity(right_speed)

left_rear_motor.setVelocity(left_speed)

right_rear_motor.setVelocity(right_speed)
```

6.5 Test Scenarios and Results

To evaluate the performance of the robot, different test scenarios were designed in the simulation environment. These tests analyze the robot's ability to follow the path and maintain stable headings under various conditions

Parameter	Results
Heading Maintainance	Acuraccy
Obstacle detection	Reliable
Obstacle Avoidance	Effective
Real-Time Telemetry	Good
Tilt Detection	Reliable
Overall Performance	Good

IMPORTABLE EXTERNPROTO

- >  WorldInfo
- >  Viewpoint
- >  TexturedBackground
- >  TexturedBackgroundLight
- >  RectangleArena "rectangle arena"
- >  DEF OBSTACLE1 Solid
- >  DEF OBSTACLE2 Solid
- >  DEF OBSTACLE3 Solid
- >  DEF EPUCK E-puck

Console - All

INFO: e-puck_line_demo: Starting controller: C:\Users\HP\Downloads\RT_project\controllers\e-puck_line_demo\e-puck_line_demo.exe
Switching to SIMULATION and resetting all BB variables.

OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0
OAM 0 side -1	LLM 0 inhibitA 0	OFM 0	LEM 0 state 0 oam_reset 0

Course Outcome

CO 1	Implemented differential drive kinematics for the autonomous mobile robot to achieve accurate line-following movement and controlled turning within the simulated environment.
CO 2	Developed and executed robot control logic for autonomous navigation, line tracking, and obstacle avoidance operations using sensor-based programming in the simulation environment.
CO 3	Integrated ground sensors and distance sensors with the robot controller to enable real-time path detection, obstacle sensing, and intelligent navigation decisions.
CO 4	Evaluated autonomous robotic control strategies including line-following algorithms and obstacle avoidance mechanisms to improve navigation efficiency and robot performance.

Level of Course Outcomes Mapped

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PSO 1	PSO 2	PSO 3
CO 1	3				3			3	2	2	1	2	2	2
CO 2	3				3			3	2	2	1	2	2	2
CO 3		3			3			3	2	2	1	2	2	2
CO 4		3			3		2	3	2	2	1	2	2	2
	3	3			3		3				1	2	2	2

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